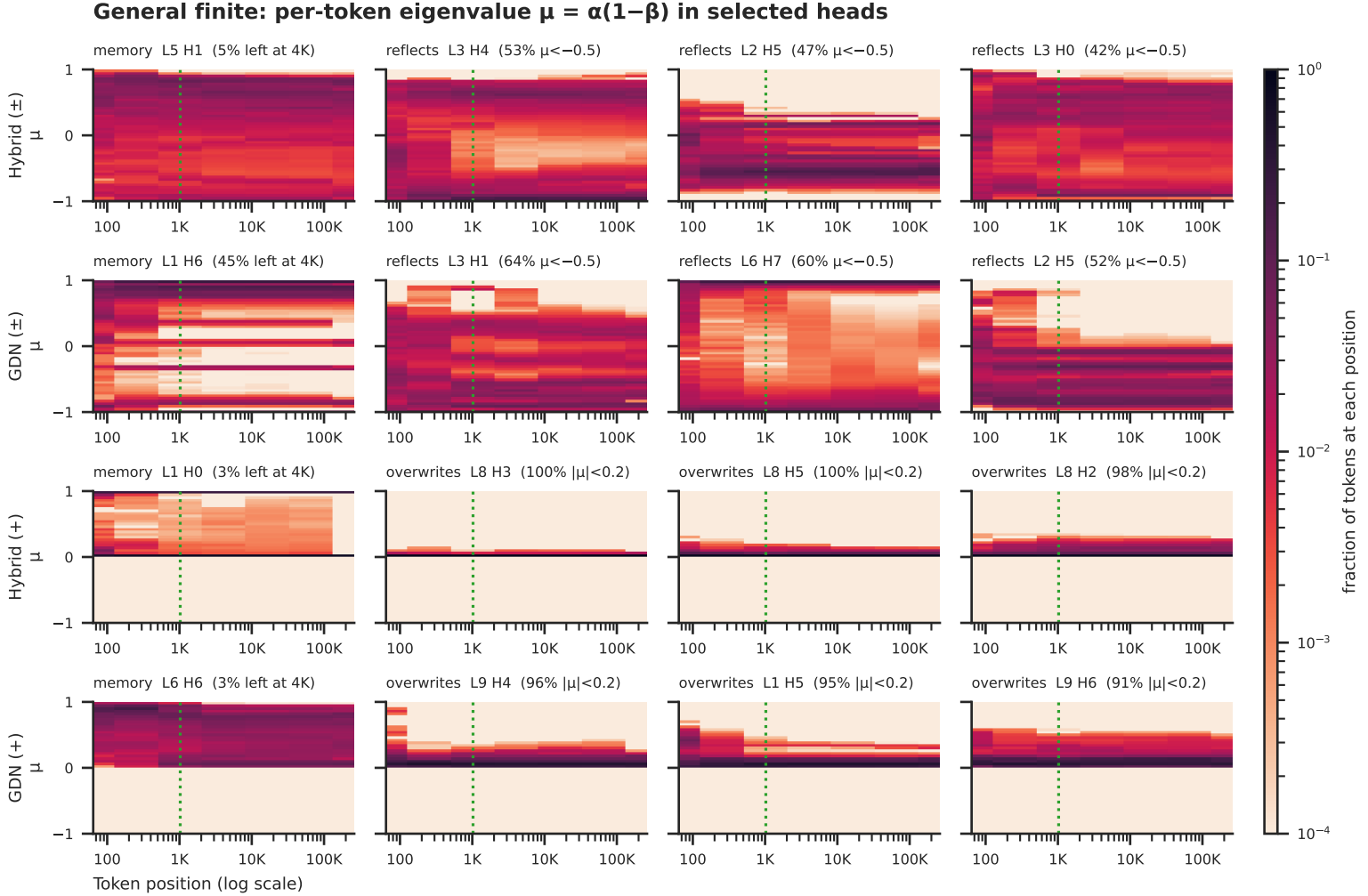

General finite: per-token eigenvalue μ = α(1−β) in selected heads
memory L5 H1 (5% left at 4K)
reflects L3 H4 (53% μ<−0.5)
reflects L2 H5 (47% μ<−0.5)
reflects L3 H0 (42% μ<−0.5)
Hybrid (±) μ
memory L1 H6 (45% left at 4K)
reflects L3 H1 (64% μ<−0.5)
reflects L6 H7 (60% μ<−0.5)
reflects L2 H5 (52% μ<−0.5)
GDN (±) μ
memory L1 H0 (3% left at 4K)
overwrites L8 H3 (100% |μ|<0.2)
overwrites L8 H5 (100% |μ|<0.2)
overwrites L8 H2 (98% |μ|<0.2)
Hybrid (+) μ
memory L6 H6 (3% left at 4K)
overwrites L9 H4 (96% |μ|<0.2)
overwrites L1 H5 (95% |μ|<0.2)
overwrites L9 H6 (91% |μ|<0.2)
GDN (+) μ
1
0
−1
100
1K
10K
100K
Token position (log scale)
fraction of tokens at each position
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}